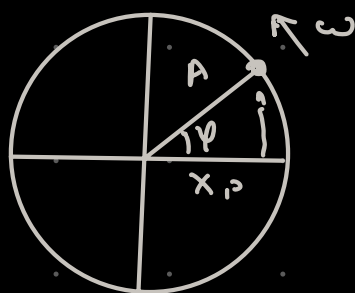


Movimento Harmônico Simple

sox q
→ actido
↓
molo

$$a = -\omega^2 \Delta x$$

O MHS pode ser representado como um.
projeção do MCV



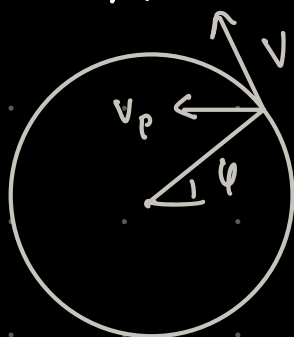
$$x_p = A \cos(\psi)$$

$$\psi = \psi_0 + \omega \Delta T$$

$$x_p = A \cos(\psi_0 + \omega \Delta T)$$

Equação horário da posição

Se $V = \omega A$



$$v_p = -V \sin(\psi)$$

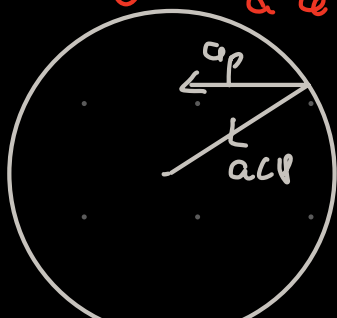
$$v_p = -\omega A \sin(\psi_0 + \omega \Delta T)$$

Eq horário
de v

$$v_{pmax} = \omega A$$

↳ x = 0 → no equilíbrio

Equação horário da aceleração



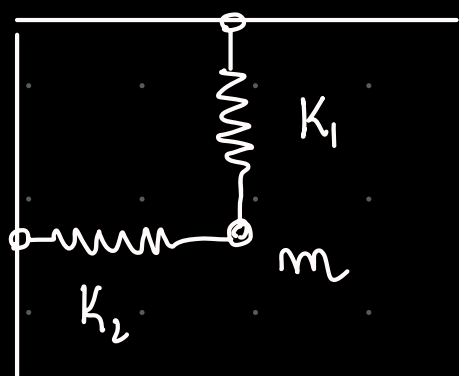
$$a_p = a_{cp} \cos(\psi)$$

$$a_p = -\omega^2 A \cos(\omega t)$$

$$a_p = -\omega^2 A (\omega \Delta \tau + \phi_0)$$

$$F = m a_p \Rightarrow F = -m\omega^2 x \quad \text{Lei de Hooke} \quad a_p = -\omega^2 x_p$$

Composição do MHS



Ex 1 $\omega_x = \omega_y = \omega$

$\phi_{0x} = \phi_{0y} = 0$

$$x = A_x \cos(\omega t)$$

$$y = A_y \cos(\omega t)$$

$$\frac{x}{A_x} = \frac{y}{A_y}$$

$$y = \underbrace{\frac{A_y}{A_x} x}_{r, \tau \Rightarrow}$$

Ex 2 $\omega_x = \omega_y = \omega$

$\phi_{0x} = \phi_{0y} + \frac{\pi}{2}$

$$x = A_x \cos(\omega t + \phi_{0y} + \frac{\pi}{2}) = -A_x \sin(\omega t + \phi_{0y})$$

$$y = A_y \cos(\omega t + \phi_{0y})$$

$$\frac{x^2}{A_x^2} + \frac{y^2}{A_y^2} = 1 \quad \text{hiperbola}$$

Dinâmica do MHS

$$a = -\omega^2 x \Rightarrow F = -kx \text{ sendo } k = m\omega^2$$

$$\sqrt{\frac{k}{m}} = \omega \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{k}{m}} \Rightarrow \boxed{T = 2\pi \sqrt{\frac{m}{k}}}$$

Energia no MHS

$$\frac{x^2}{A^2} + \frac{v^2}{\omega^2 A^2} = 1 \Rightarrow \left\{ \frac{kx^2}{2} + \frac{mv^2}{2} = \frac{kA^2}{2} \right\}$$
$$\frac{x^2}{A^2} + \frac{v^2 m}{kA^2} = 1 \quad \swarrow$$

$$E_c = \frac{mv^2}{2} \quad E_p = \frac{kx^2}{2} \quad E_m = \frac{kA^2}{2}$$